

AISSAOUI Ismail

State Engineer in Artificial Intelligence & Data Science

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RESEARCH SUMMARY

State Engineer (ESI-SBA) specializing in **Physics-Informed Machine Learning (PINN)** and **Computer Vision**. Expert in industrial digital twins, real-time monitoring, and the integration of physical laws into deep learning architectures. PhD candidate focused on sustainable energy systems.

EDUCATION

National School of Computer Science (ESI-SBA, Algeria) 2021 – 2026
State Engineer's Degree & MSc in Artificial Intelligence

RESEARCH PROJECTS

- Physics-Informed Digital Twin — Sonatrach** 2026
Development of a generative surrogate for gas turbine monitoring.
- Implemented custom **PINN loss functions** to enforce thermodynamic laws.
 - Developed a dual-track architecture using **Mamba-SSM** and **xLSTM**.
 - Optimized for resource-constrained edge devices with **0.04ms** latency.
- Industrial Safety & CV Monitoring (YOLOv8)** 2025
Computerized image processing system for safety equipment detection and tracking. *Key skill: Computerized Image Processing & Object Tracking.*
- Recognizini — Real-Time Face Recognition** 2025
High-speed vision system with active learning and incremental embeddings.
- Geo-RAG — Semantic Knowledge Retrieval** 2025
Knowledge extraction from multi-source industrial and geospatial data.
- Career Connect — Structural GNN Intelligence** 2024 – 2025
Relational modeling of complex systems via Graph Neural Networks.

TECHNICAL SKILLS

- Physics-Informed AI:** PINN (Loss design), Koopman Theory, Physics-Aware Loss.
- Computer Vision:** YOLOv8, OpenCV, Image Segmentation, Active Learning.
- Sequence Modeling:** Mamba-SSM, xLSTM, Informer, TCN.
- Engineering:** PyTorch, C++, Docker, MLflow, CUDA, Linux, Git.

LANGUAGES

Arabic (Native), English (Advanced - Professional Research Level), French (Fluent).